Worksheet 6: Universality and Quantum Circuits II

Solutions

Quantum Computation and Error Correction
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Discussed on October 27, 2026.

Same conventions as on the question sheet: qubits are numbered from 0, $|ab\rangle$ has qubit 0 in state a, $P_0 = |0\rangle\langle 0| = \frac{1}{2}(\mathbb{I} + Z)$, $P_1 = |1\rangle\langle 1| = \frac{1}{2}(\mathbb{I} - Z)$, and a circuit "first G_1, then G_2" is the matrix $G_2 G_1$. $R_{\hat{n}}(\theta) = \exp(-\frac{i\theta}{2}\,\hat{n}\cdot\vec{\sigma})$. I will write $\omega = \mathrm{e}^{i\pi/4}$ for the phase in $T = \mathrm{diag}(1, \omega)$, it saves a lot of exponents. If you find a mistake in these solutions, which is entirely possible, please tell me.

Part A — Single-qubit gates and rotations

1. Rotations of a single qubit

(a) We first need $(\hat{n}\cdot\vec{\sigma})^2 = \mathbb{I}$. Just expand:

$$(\hat{n}\cdot\vec{\sigma})^2 = \sum_j n_j^2\,\sigma_j^2 + \sum_{j<k} n_j n_k\,\{\sigma_j, \sigma_k\} = \Big(\sum_j n_j^2\Big)\mathbb{I} + 0 = \mathbb{I},$$

using $\sigma_j^2 = \mathbb{I}$, $\{\sigma_j, \sigma_k\} = 0$ for $j \neq k$, and $|\hat{n}| = 1$. So even powers of $\hat{n}\cdot\vec{\sigma}$ are $\mathbb{I}$ and odd powers are $\hat{n}\cdot\vec{\sigma}$ itself. Now split the exponential series into even and odd k:

$$\exp\Big(-\tfrac{i\theta}{2}\,\hat{n}\cdot\vec{\sigma}\Big) = \sum_{k\ \text{even}} \frac{(-i\theta/2)^k}{k!}\,\mathbb{I} + \sum_{k\ \text{odd}} \frac{(-i\theta/2)^k}{k!}\,\hat{n}\cdot\vec{\sigma} = \cos\tfrac{\theta}{2}\,\mathbb{I} - i\sin\tfrac{\theta}{2}\,\hat{n}\cdot\vec{\sigma}.$$

The two sums are the even and odd parts of e^{ix} at $x = -\theta/2$, i.e. $\cos x$ and $i\sin x$. So this is nothing but Euler's formula, with the number i replaced by the matrix $-i\,\hat{n}\cdot\vec{\sigma}$, which also squares to $-\mathbb{I}$. Almost everything we do with single-qubit gates below comes back to this one line.

(b) Put $\hat{n} = \hat{z}$. Then $\hat{n}\cdot\vec{\sigma} = Z = \mathrm{diag}(1, -1)$ and

$$R_z(\theta) = \cos\tfrac{\theta}{2}\,\mathbb{I} - i\sin\tfrac{\theta}{2}\,Z = \begin{pmatrix} \mathrm{e}^{-i\theta/2} & 0 \\ 0 & \mathrm{e}^{i\theta/2} \end{pmatrix} = \mathrm{e}^{-i\theta/2}\begin{pmatrix} 1 & 0 \\ 0 & \mathrm{e}^{i\theta} \end{pmatrix}.$$

Read the last equality backwards: $\mathrm{diag}(1, \mathrm{e}^{i\theta}) = \mathrm{e}^{i\theta/2} R_z(\theta)$. Since Z, S, T are $\mathrm{diag}(1, \mathrm{e}^{i\theta})$ with $\theta = \pi, \pi/2, \pi/4$,

$$Z = \mathrm{e}^{i\pi/2} R_z(\pi), \qquad S = \mathrm{e}^{i\pi/4} R_z(\pi/2), \qquad T = \mathrm{e}^{i\pi/8} R_z(\pi/4),$$

and the phase factors are i, $\mathrm{e}^{i\pi/4}$, $\mathrm{e}^{i\pi/8}$. Each gate is the square root of the previous one, so the angle halves and the phase halves. Nothing deep.

(c) With $\hat{n} = \hat{y}$,

$$R_y(\pi/2) = \cos\tfrac{\pi}{4}\,\mathbb{I} - i\sin\tfrac{\pi}{4}\,Y = \frac{1}{\sqrt{2}}\left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - i\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\right] = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

That is not H yet; the sign in the top right is wrong. Multiplying by Z on the right flips the sign of the second column and fixes exactly that:

$$R_y(\pi/2)\,Z = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = H,$$

with no phase at all. Then $Z = \mathrm{e}^{i\pi/2} R_z(\pi)$ from (b) gives $H = \mathrm{e}^{i\pi/2} R_y(\pi/2) R_z(\pi)$.

A common slip here is to think $R_y(\pi/2)$ alone is the Hadamard because it sends $|0\rangle$ to $|+\rangle$. It does, but it sends $|1\rangle$ to $-|-\rangle$, and that relative sign matters.

(d) Every probability you can ever measure has the form $|\langle\phi|\psi'\rangle|^2$ for the output state $|\psi'\rangle$ and some outcome $|\phi\rangle$. If $|\psi'\rangle = \mathrm{e}^{i\alpha} U|\psi\rangle$, then $|\langle\phi|\psi'\rangle|^2 = |\mathrm{e}^{i\alpha}|^2 |\langle\phi|U|\psi\rangle|^2 = |\langle\phi|U|\psi\rangle|^2$. The α has disappeared, so no experiment can tell $\mathrm{e}^{i\alpha}U$ and U apart, and it makes sense to call them the same gate. Keep in mind though that this only holds for the gate on its own. Once we apply U conditionally on another qubit (Q4(c)), the phase is applied conditionally too, and a conditional phase is a relative phase. Those are measurable.

2. The Z–Y–Z decomposition

(a) Let $c = \cos\frac{\gamma}{2}$ and $s = \sin\frac{\gamma}{2}$. From Q1, $R_z(\beta) = \mathrm{diag}(\mathrm{e}^{-i\beta/2}, \mathrm{e}^{i\beta/2})$ and $R_y(\gamma) = c\,\mathbb{I} - isY = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}$. Multiplying by a diagonal matrix on the left rescales the rows, on the right it rescales the columns, so

$$R_z(\beta) R_y(\gamma) R_z(\delta) = \begin{pmatrix} \mathrm{e}^{-i\beta/2} c & -\mathrm{e}^{-i\beta/2} s \\ \mathrm{e}^{i\beta/2} s & \mathrm{e}^{i\beta/2} c \end{pmatrix} \begin{pmatrix} \mathrm{e}^{-i\delta/2} & 0 \\ 0 & \mathrm{e}^{i\delta/2} \end{pmatrix} = \begin{pmatrix} \mathrm{e}^{-i(\beta+\delta)/2} c & -\mathrm{e}^{-i(\beta-\delta)/2} s \\ \mathrm{e}^{i(\beta-\delta)/2} s & \mathrm{e}^{i(\beta+\delta)/2} c \end{pmatrix}.$$

(b) Take $U = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$ with orthonormal columns, so $|a|^2 + |b|^2 = 1$, $|c|^2 + |d|^2 = 1$ and $\bar{a}c + \bar{b}d = 0$.

The third condition says $(c, d) \perp (a, b)$ in $\mathbb{C}^2$. In two dimensions the orthogonal complement of a non-zero vector is a line, and the vector $(-\bar{b}, \bar{a})$ lies on it (check: $\bar{a}(-\bar{b}) + \bar{b}\,\bar{a} = 0$). So $(c, d) = \lambda(-\bar{b}, \bar{a})$ for some complex λ, and $|c|^2 + |d|^2 = 1$ forces $|\lambda| = 1$, say $\lambda = \mathrm{e}^{i\phi}$. Now pull half the phase out of the whole matrix:

$$U = \begin{pmatrix} a & -\mathrm{e}^{i\phi}\bar{b} \\ b & \mathrm{e}^{i\phi}\bar{a} \end{pmatrix} = \mathrm{e}^{i\phi/2} \begin{pmatrix} a\mathrm{e}^{-i\phi/2} & -\overline{b\mathrm{e}^{-i\phi/2}} \\ b\mathrm{e}^{-i\phi/2} & \overline{a\mathrm{e}^{-i\phi/2}} \end{pmatrix} = \mathrm{e}^{i\alpha} \begin{pmatrix} a' & -\bar{b}' \\ b' & \bar{a}' \end{pmatrix},$$

with $\alpha = \phi/2$, $a' = a\mathrm{e}^{-i\phi/2}$, $b' = b\mathrm{e}^{-i\phi/2}$, and clearly $|a'|^2 + |b'|^2 = 1$. The bracketed matrix has determinant 1, so what we have shown is that any 2×2 unitary is a phase times an $SU(2)$ matrix.

(c) Compare the two matrices entry by entry. The modulus of the top-left entry gives $|a'| = \cos\frac{\gamma}{2}$; if we agree to take $\gamma \in [0, \pi]$ then $\cos\frac{\gamma}{2}$ and $\sin\frac{\gamma}{2}$ are both non-negative, so $\gamma = 2\arccos|a'|$, and then $|b'| = \sin\frac{\gamma}{2}$ comes out automatically from $|a'|^2 + |b'|^2 = 1$. For the phases, $\arg a' = -\frac{\beta+\delta}{2}$ and $\arg b' = \frac{\beta-\delta}{2}$, which we solve as

$$\beta = \arg b' - \arg a', \qquad \delta = -\arg a' - \arg b'.$$

With these values, $\mathrm{e}^{i\alpha}$ times the matrix of (a) matches U entry for entry, so

$$U = \mathrm{e}^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta),$$

which is the claim.

(If $\gamma = 0$ or π then b' or a' is zero and has no argument. Then only $\beta + \delta$, or only $\beta - \delta$, is determined, and you can pick the other combination freely. So the decomposition always exists but it is not unique; this comes up again in (d).)

(d) Q1(c) already says $H = \mathrm{e}^{i\pi/2} R_y(\pi/2) R_z(\pi)$, and we can insert $R_z(0) = \mathbb{I}$ on the left, so

$$(\alpha, \beta, \gamma, \delta) = (\pi/2,\ 0,\ \pi/2,\ \pi).$$

As a check that the recipe of (c) gives the same: H has $a = b = c = 1/\sqrt{2}$, $d = -1/\sqrt{2}$, so $(c, d) = \mathrm{e}^{i\phi}(-\bar{b}, \bar{a})$ needs $\mathrm{e}^{i\phi} = -1$, i.e. $\phi = \pi$, $\alpha = \pi/2$ and $a' = b' = \mathrm{e}^{-i\pi/2}/\sqrt{2}$. Then $\gamma = 2\arccos(1/\sqrt{2}) = \pi/2$, $\beta = \arg b' - \arg a' = 0$ and $\delta = -\arg a' - \arg b' = \pi$. Same thing.

3. A discrete gate set: H and T

(a) With $T = \mathrm{diag}(1, \omega)$,

$$TXT^\dagger = \begin{pmatrix}1&0\\0&\omega\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}1&0\\0&\bar\omega\end{pmatrix} = \begin{pmatrix}0&\bar\omega\\\omega&0\end{pmatrix} = \cos\tfrac{\pi}{4}\begin{pmatrix}0&1\\1&0\end{pmatrix} + \sin\tfrac{\pi}{4}\begin{pmatrix}0&-i\\i&0\end{pmatrix} = \frac{X+Y}{\sqrt{2}},$$

where the third equality is just $\omega = \frac{1}{\sqrt{2}}(1+i)$ and $\bar\omega = \frac{1}{\sqrt{2}}(1-i)$. So T sends X to an honest superposition of two Paulis rather than to $\pm$ one Pauli, and by the Worksheet 5 definition it is not a Clifford gate. On the Bloch sphere this is obvious: T is a 45° turn about $\hat z$, which takes the $\hat x$ axis halfway towards $\hat y$, whereas a Clifford has to permute the six points $\pm\hat x, \pm\hat y, \pm\hat z$ among themselves.

(b) Write $T = \mathrm{e}^{i\pi/8} R_z(\pi/4) = \mathrm{e}^{i\pi/8}\exp(-i\tfrac{\pi}{8}Z)$. Because $H^2 = \mathbb{I}$, we have $HM^kH = (HMH)^k$ for every k, and therefore $H\mathrm{e}^{M}H = \mathrm{e}^{HMH}$ term by term in the series. With $HZH = X$ this gives

$$HTH = \mathrm{e}^{i\pi/8}\, H\exp(-i\tfrac{\pi}{8}Z)H = \mathrm{e}^{i\pi/8}\exp(-i\tfrac{\pi}{8}X) = \mathrm{e}^{i\pi/8}R_x(\pi/4).$$

(c) By (b), $THTH = T\,(HTH) = \mathrm{e}^{i\pi/8}R_z(\pi/4)\cdot\mathrm{e}^{i\pi/8}R_x(\pi/4) = \mathrm{e}^{i\pi/4}R_z(\pi/4)R_x(\pi/4)$. The two rotations have determinant 1 and so does their product, and Q2(b) tells us a determinant-one unitary looks like $\cos\frac{\theta}{2}\mathbb{I} - i\sin\frac{\theta}{2}\,\hat m\cdot\vec\sigma$ for some unit vector $\hat m$, which is $R_{\hat m}(\theta)$. Its trace is $2\cos\frac{\theta}{2}$, the Paulis being traceless. On the other hand, with $c = \cos\frac{\pi}{8}$, $s = \sin\frac{\pi}{8}$,

$$R_z(\pi/4)R_x(\pi/4) = (c\mathbb{I} - isZ)(c\mathbb{I} - isX) = c^2\,\mathbb{I} - ics\,(Z+X) - s^2\,ZX,$$

and X, Z, $ZX = iY$ all have zero trace, so $\mathrm{tr} = 2c^2$. Equating,

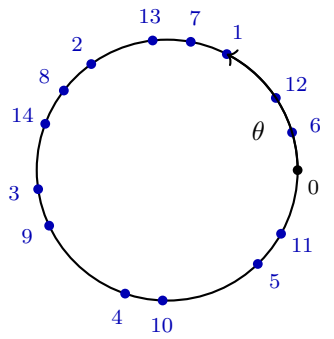
$$\cos\tfrac{\theta}{2} = \cos^2\tfrac{\pi}{8} = \tfrac{1+\cos(\pi/4)}{2} = \tfrac{2+\sqrt2}{4} \approx 0.854, \qquad \text{so} \quad \theta \approx 1.096 \text{ rad} \approx 62.8°.$$

We were not asked for the axis, but it is sitting right there: $-i\sin\frac{\theta}{2}\,\hat m\cdot\vec\sigma = -ics(X+Z) - is^2Y$ gives $\hat m \propto (\cos\frac{\pi}{8},\ \sin\frac{\pi}{8},\ \cos\frac{\pi}{8})$.

(d) Since $R_{\hat m}(\theta)^k = R_{\hat m}(k\theta)$, the k-th power of $THTH$ is, up to a phase $\mathrm{e}^{ik\pi/4}$ that we do not care about, a rotation about $\hat m$ by $k\theta$. So the question is which angles $k\theta$ mod 2π we can reach. Picture them as points on a circle.

First, no two of them coincide. If $k\theta \equiv k'\theta \bmod 2\pi$ with $k \neq k'$ then $(k-k')\theta = 2\pi n$ for some integer n, which would make $\theta/\pi = 2n/(k-k')$ rational. So the points $\theta, 2\theta, 3\theta, \ldots$ are all different.

Infinitely many distinct points on a circle have to crowd together somewhere, and from that we get everywhere. Chop the circle into M arcs of length $2\pi/M$. Of the first $M+1$ points, two land in the same arc, say $k_1\theta$ and $k_2\theta$ with $k_2 > k_1$. Then $d = (k_2 - k_1)\theta \bmod 2\pi$ is within $2\pi/M$ of zero, and it is not zero by the previous paragraph. The multiples $d, 2d, 3d, \ldots$ are again of the form $k\theta$, and they march around the circle in steps shorter than $2\pi/M$, so they cannot skip any arc. Hence for any ϕ there is a k with $k\theta$ within $2\pi/M$ of ϕ. Since M was arbitrary, some power of $THTH$ gets as close as we like to $R_{\hat m}(\phi)$, whatever ϕ is.



The first fourteen multiples of $\theta \approx 62.8°$ on the circle. No repeats, and the gaps keep shrinking.

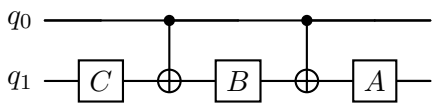
(e) Look back at the proof in Q2: nowhere did it matter that the two axes were $\hat{z}$ and $\hat{y}$. If you rotate your coordinate system so that $\hat{m}$ points along $\hat{z}$ and redo the algebra with the second axis $\hat{n}$ in place of $\hat{y}$ (slightly messier, but the same idea; it is Exercise 4.11 in Nielsen & Chuang), you get $U = \mathrm{e}^{i\alpha} R_{\hat{m}}(\beta) R_{\hat{n}}(\gamma) R_{\hat{m}}(\delta)$ for any two axes that are not parallel. Part (d) gives us rotations about $\hat{m}$ to any accuracy. For a second axis, conjugate by H: $H(THTH)H = HTHT$ is a rotation by the same θ about $\hat{m}' = H\hat{m}H$, which means swapping the x and z components of $\hat{m}$ and flipping the sign of y, so $\hat{m}' \propto (\cos\frac{\pi}{8}, -\sin\frac{\pi}{8}, \cos\frac{\pi}{8})$. Not parallel to $\hat{m}$, good. Now approximate each of $R_{\hat{m}}(\beta)$, $R_{\hat{m}'}(\gamma)$, $R_{\hat{m}}(\delta)$ by a suitable power of $THTH$ or $HTHT$; the three errors add up (that is Q7) but each can be made as small as we want. So every single-qubit gate is approximable with H and T alone. The result from Q2 that we are leaning on is the three-rotation decomposition, in its "any two axes" form.

It is worth pausing on how little went into this. The whole universality of $\{H, T\}$ for one qubit hangs on the single fact that the angle in (c) is an irrational multiple of π.

Part B — Controlled gates

4. Building controlled-U from cx and single-qubit gates

(a) The circuit in question is



and as a matrix, reading right to left, it is $(\mathbb{I} \otimes A)\,\mathtt{cx}\,(\mathbb{I} \otimes B)\,\mathtt{cx}\,(\mathbb{I} \otimes C)$. It is enough to see what it does when the control is $|0\rangle$ and when it is $|1\rangle$; linearity does the rest.

If the control is $|0\rangle$, both cx's do nothing, and the target just sees C, then B, then A. So $|0\rangle \otimes |\psi\rangle \mapsto |0\rangle \otimes ABC|\psi\rangle = |0\rangle \otimes |\psi\rangle$.

If the control is $|1\rangle$, each cx acts as X on the target, which therefore sees C, X, B, X, A in that order: $|1\rangle \otimes |\psi\rangle \mapsto |1\rangle \otimes AXBXC|\psi\rangle = |1\rangle \otimes U|\psi\rangle$.

Putting the two together, the circuit is $P_0 \otimes \mathbb{I} + P_1 \otimes U$, which is controlled-$U$ with qubit 0 as the control.

(b) By Q1(a), $R_z(\theta) = c\mathbb{I} - isZ$ with $c = \cos\frac{\theta}{2}$, $s = \sin\frac{\theta}{2}$, and $XZX = -Z$. So

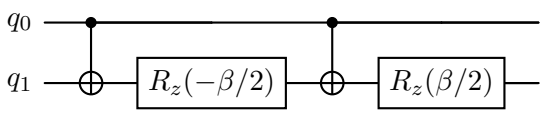
$$XR_z(\theta)X = c\,\mathbb{I} - is\,XZX = c\,\mathbb{I} + is\,Z = R_z(-\theta),$$

and the same argument with $XYX = -Y$ gives $XR_y(\theta)X = R_y(-\theta)$. Geometrically, X is a half turn about $\hat{x}$; it turns the $\hat{y}$ and $\hat{z}$ axes upside down, so a rotation about either of them changes its sense.

Now take $A = R_z(\beta/2)$, $B = R_z(-\beta/2)$, $C = \mathbb{I}$. Then $ABC = R_z(\beta/2)R_z(-\beta/2) = \mathbb{I}$, and

$$AXBXC = R_z(\beta/2)\left[XR_z(-\beta/2)X\right] = R_z(\beta/2)R_z(\beta/2) = R_z(\beta),$$

so by (a) the circuit

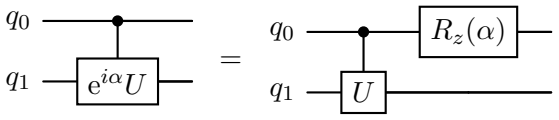


is controlled-$R_z(\beta)$.

(c) Straight from the definition,

$$\text{controlled-}(\mathrm{e}^{i\alpha}U) = P_0 \otimes \mathbb{I} + P_1 \otimes \mathrm{e}^{i\alpha}U = \big[(P_0 + \mathrm{e}^{i\alpha}P_1) \otimes \mathbb{I}\big]\big(P_0 \otimes \mathbb{I} + P_1 \otimes U\big).$$

To see the second equality, multiply out and use $P_0P_1 = 0$, $P_0^2 = P_0$, $P_1^2 = P_1$: the cross terms die and you are left with $P_0 \otimes \mathbb{I} + \mathrm{e}^{i\alpha}P_1 \otimes U$. The extra factor is $\mathrm{diag}(1, \mathrm{e}^{i\alpha})$ on the control, and by Q1(b) that is $\mathrm{e}^{i\alpha/2}R_z(\alpha)$. (The two factors commute, so it does not matter whether we say the phase gate comes before or after the controlled-U.) As circuits,



up to the global phase $\mathrm{e}^{i\alpha/2}$.

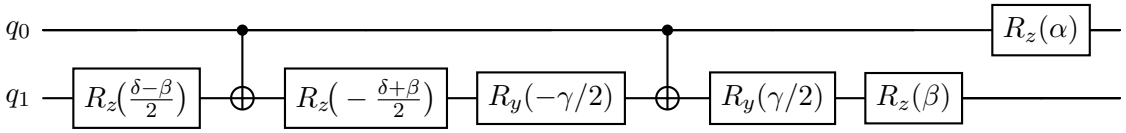
So a phase that was invisible on U alone has turned into a perfectly ordinary z-rotation on the control. If you want to see it in an experiment: prepare the control in $|+\rangle$ and the target in an eigenstate $|u\rangle$ of U with eigenvalue 1. Controlled-U leaves $|+\rangle|u\rangle$ alone, whereas controlled-$(\mathrm{e}^{i\alpha}U)$ turns it into $\frac{1}{\sqrt{2}}(|0\rangle + \mathrm{e}^{i\alpha}|1\rangle)|u\rangle$, and measuring the control in the $\pm$ basis now gives $|-\rangle$ with probability $\sin^2\frac{\alpha}{2}$ instead of zero. (This is also how phase estimation works, if you have seen it.)

(d) Take $A = R_z(\beta)R_y(\gamma/2)$, $B = R_y(-\gamma/2)R_z\big(-\frac{\delta+\beta}{2}\big)$ and $C = R_z\big(\frac{\delta-\beta}{2}\big)$. Then

$$ABC = R_z(\beta)\underbrace{R_y(\gamma/2)R_y(-\gamma/2)}_{\mathbb{I}}R_z\big(-\tfrac{\delta+\beta}{2}\big)R_z\big(\tfrac{\delta-\beta}{2}\big) = R_z(\beta)R_z(-\beta) = \mathbb{I},$$

$$\begin{aligned} AXBXC &= R_z(\beta)R_y(\gamma/2)\,\underbrace{XR_y(-\gamma/2)X}_{R_y(\gamma/2)}\,\underbrace{XR_z\big(-\tfrac{\delta+\beta}{2}\big)X}_{R_z(\frac{\delta+\beta}{2})}\,R_z\big(\tfrac{\delta-\beta}{2}\big) \\ &= R_z(\beta)\,R_y(\gamma)\,R_z\big(\tfrac{\delta+\beta}{2} + \tfrac{\delta-\beta}{2}\big) = R_z(\beta)R_y(\gamma)R_z(\delta), \end{aligned}$$

where in the second line I slipped an $X^2 = \mathbb{I}$ in between the two factors of B so that (b) could be used twice. So the circuit of (a) gives controlled-$[R_z(\beta)R_y(\gamma)R_z(\delta)]$, and by (c) the missing $\mathrm{e}^{i\alpha}$ is an $R_z(\alpha)$ on the control. All together, for $U = \mathrm{e}^{i\alpha}R_z(\beta)R_y(\gamma)R_z(\delta)$,



where the two boxes between the `cx`'s are B written out (earlier factor on the left) and the two after the second `cx` are A. Two `cx` gates, no matter what U is. Combined with Q2 and Q3 this means $\{H, T, \mathtt{cx}\}$ can do every controlled gate.

5. The Toffoli gate needs T

(a) It helps to write the Toffoli with qubit 0 singled out: $\mathtt{ccx} = P_0 \otimes \mathbb{I} \otimes \mathbb{I} + P_1 \otimes \mathtt{cx}(1,2)$, that is, "if qubit 0 is 1, do $\mathtt{cx}(1,2)$". Let $X_0 = X \otimes \mathbb{I} \otimes \mathbb{I}$. Since $XP_0 = P_1X$ and $XP_1 = P_0X$, pushing X_0 through from the left swaps the projectors:

$$X_0\,\mathtt{ccx} = (XP_0) \otimes \mathbb{I} + (XP_1) \otimes \mathtt{cx}(1,2) = \big[P_1 \otimes \mathbb{I} + P_0 \otimes \mathtt{cx}(1,2)\big]X_0.$$

Now multiply by ccx on the left, remembering $P_0P_1 = 0$, $P_a^2 = P_a$ and $\mathtt{cx}(1,2)^2 = \mathbb{I}$:

$$\mathtt{ccx}\big[P_1 \otimes \mathbb{I} + P_0 \otimes \mathtt{cx}(1,2)\big] = P_0 \otimes \mathtt{cx}(1,2) + P_1 \otimes \mathtt{cx}(1,2) = \mathbb{I} \otimes \mathtt{cx}(1,2).$$

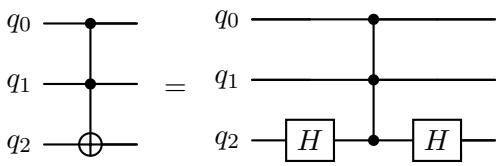
So $\mathtt{ccx}\, X_0\, \mathtt{ccx} = \mathtt{cx}(1,2)\, X_0 = X_0\, \mathtt{cx}(1,2)$; the two factors live on different qubits so their order is irrelevant. Is that a Pauli product? No: $\mathtt{cx}(1,2) = P_0 \otimes \mathbb{I} + P_1 \otimes X = \frac{1}{2}(\mathbb{I} \otimes \mathbb{I} + Z \otimes \mathbb{I} + \mathbb{I} \otimes X - Z \otimes X)$ is a sum of four different Pauli products with equal weights, and Pauli products are linearly independent, so this is not $\pm$ a single one. A Clifford would have to send X_0 to $\pm$ a Pauli product. The Toffoli does not, so it is not Clifford.

In the language of Worksheet 5: a bit flip on the top control, pushed through a Toffoli, comes out as a bit flip *and* a cx on the other two qubits. That is no longer a Pauli error and the simple error bookkeeping breaks down.

(b) Both gates are the identity except on $|111\rangle$. Concretely $\mathtt{ccz} = \mathbb{I} - 2P_1 \otimes P_1 \otimes P_1$, since it flips the sign of $|111\rangle$ and nothing else, while $\mathtt{ccx} = \mathbb{I} + P_1 \otimes P_1 \otimes (X - \mathbb{I})$, since on $|11\rangle \otimes |c\rangle$ it replaces the identity on the last qubit by X. Conjugate ccz by H on qubit 2 and use $HP_1H = |-\rangle\langle -|$:

$$(\mathbb{I} \otimes \mathbb{I} \otimes H)\, \mathtt{ccz}\, (\mathbb{I} \otimes \mathbb{I} \otimes H) = \mathbb{I} - 2P_1 \otimes P_1 \otimes |-\rangle\langle -|.$$

But $X - \mathbb{I} = -2|-\rangle\langle -|$, because $X - \mathbb{I}$ kills $|+\rangle$ and multiplies $|-\rangle$ by -2. So the right-hand side is ccx. Drawn out,



which is the same story as $\mathtt{cx} = (\mathbb{I} \otimes H)\, \mathtt{cz}\, (\mathbb{I} \otimes H)$ on Worksheet 5: ccz is diagonal in the Z basis of the last qubit, and H swaps that for the X basis.

(c) On a basis state $Z_j|abc\rangle = (-1)^{\text{bit } j}|abc\rangle$, so

$$\tfrac{1}{2}(\mathbb{I} - Z_j)|abc\rangle = \tfrac{1}{2}\big(1 - (-1)^{\text{bit } j}\big)|abc\rangle = (\text{bit } j)|abc\rangle;$$

the bracket is $\frac{1}{2}(1-1) = 0$ if the bit is 0 and $\frac{1}{2}(1+1) = 1$ if it is 1. The three operators $\frac{1}{2}(\mathbb{I} - Z_j)$ are diagonal, so they commute and their product has eigenvalue abc on $|abc\rangle$. Therefore

$$\exp\Big[i\pi \cdot \tfrac{1}{2}(\mathbb{I} - Z_0) \cdot \tfrac{1}{2}(\mathbb{I} - Z_1) \cdot \tfrac{1}{2}(\mathbb{I} - Z_2)\Big]|abc\rangle = \mathrm{e}^{i\pi abc}|abc\rangle = (-1)^{abc}|abc\rangle,$$

and an operator that agrees with ccz on every basis state is ccz. The three halves give the $\frac{1}{8}$. Multiplying out,

$$(\mathbb{I} - Z_0)(\mathbb{I} - Z_1)(\mathbb{I} - Z_2) = \mathbb{I} - Z_0 - Z_1 - Z_2 + Z_0Z_1 + Z_0Z_2 + Z_1Z_2 - Z_0Z_1Z_2,$$

which is the second form on the sheet.

(d) Everything in the exponent is diagonal, so everything commutes, and $\mathrm{e}^{A+B} = \mathrm{e}^A\mathrm{e}^B$ is allowed. The $\mathbb{I}$ term is a global phase $\mathrm{e}^{i\pi/8}$; the other seven terms give seven factors,

$$\mathtt{ccz} = \mathrm{e}^{i\pi/8} \prod_j \mathrm{e}^{-i\pi Z_j/8} \prod_{i<j} \mathrm{e}^{+i\pi Z_iZ_j/8}\ \mathrm{e}^{-i\pi Z_0Z_1Z_2/8}.$$

The single-qubit ones are T gates in disguise:

$$\mathrm{e}^{-i\pi Z/8} = \begin{pmatrix} \mathrm{e}^{-i\pi/8} & 0 \\ 0 & \mathrm{e}^{i\pi/8} \end{pmatrix} = \mathrm{e}^{-i\pi/8} \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{e}^{i\pi/4} \end{pmatrix} = \mathrm{e}^{-i\pi/8}\, T, \qquad \mathrm{e}^{+i\pi Z/8} = \mathrm{e}^{i\pi/8}\, T^\dagger.$$

For the others we use the hint. Because $\mathtt{cx}^2 = \mathbb{I}$, $\mathtt{cx}\, M^k\, \mathtt{cx} = (\mathtt{cx}\, M\, \mathtt{cx})^k$, so $\mathtt{cx}\, f(M)\, \mathtt{cx} = f(\mathtt{cx}\, M\, \mathtt{cx})$ for any power series f, in particular for $f = \exp$. Together with the Worksheet 5 rule $\mathtt{cx}(i,j)\, Z_j\, \mathtt{cx}(i,j) = Z_i Z_j$ (a Z on the target gets copied to the control) this gives

$$\mathtt{cx}(i,j)\, \mathrm{e}^{i\phi Z_j}\, \mathtt{cx}(i,j) = \mathrm{e}^{i\phi\, \mathtt{cx}(i,j) Z_j \mathtt{cx}(i,j)} = \mathrm{e}^{i\phi Z_i Z_j}.$$

Do it twice for the three-body term: $\mathtt{cx}(1,2)$ turns Z_2 into $Z_1 Z_2$, and $\mathtt{cx}(0,2)$ then leaves Z_1 alone and turns Z_2 into $Z_0 Z_2$, so

$$\mathtt{cx}(0,2)\, \mathtt{cx}(1,2)\, \mathrm{e}^{i\phi Z_2}\, \mathtt{cx}(1,2)\, \mathtt{cx}(0,2) = \mathtt{cx}(0,2)\, \mathrm{e}^{i\phi Z_1 Z_2}\, \mathtt{cx}(0,2) = \mathrm{e}^{i\phi Z_0 Z_1 Z_2}.$$

If you read the left-hand side as a circuit (time order $\mathtt{cx}(0,2)$, $\mathtt{cx}(1,2)$, phase, $\mathtt{cx}(1,2)$, $\mathtt{cx}(0,2)$) it says: compute the parity $a \oplus b \oplus c$ onto qubit 2, apply a phase depending on that parity, uncompute. A $Z_0 Z_1 Z_2$ rotation is nothing more than that.

(e) Set $\phi = \pm\pi/8$ in (d). Each of the seven factors becomes a T or $T^\dagger$ wrapped in cx's, and the loose phases cancel exactly:

$$\mathrm{e}^{i\pi/8} \cdot \left(\mathrm{e}^{-i\pi/8}\right)^3 \cdot \left(\mathrm{e}^{i\pi/8}\right)^3 \cdot \mathrm{e}^{-i\pi/8} = 1.$$

Writing T_j for T on qubit j,

$$\begin{aligned}\mathtt{ccz} = T_0 T_1 T_2 \;&\cdot\; \big[\mathtt{cx}(0,1)\, T_1^\dagger\, \mathtt{cx}(0,1)\big]\big[\mathtt{cx}(0,2)\, T_2^\dagger\, \mathtt{cx}(0,2)\big]\big[\mathtt{cx}(1,2)\, T_2^\dagger\, \mathtt{cx}(1,2)\big] \\ &\cdot\; \big[\mathtt{cx}(0,2)\mathtt{cx}(1,2)\, T_2\, \mathtt{cx}(1,2)\mathtt{cx}(0,2)\big].\end{aligned}$$

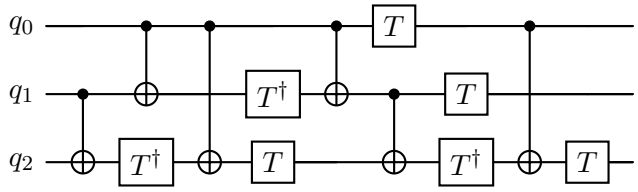
As written that is seven $T^{\pm 1}$ and eight cx. Two of the cx can be got rid of. The seven factors all commute, so we may put the two brackets that contain $\mathtt{cx}(0,2)$ next to each other, and then a $\mathtt{cx}(0,2)\mathtt{cx}(0,2)$ pair cancels. On qubit 2 this leaves, in time order,

$$\mathtt{cx}(1,2),\ T^\dagger,\ \mathtt{cx}(1,2),\ \mathtt{cx}(0,2),\ T^\dagger,\ \mathtt{cx}(1,2),\ T,\ \mathtt{cx}(1,2),\ \mathtt{cx}(0,2).$$

The two gates $\mathtt{cx}(1,2)$ and $\mathtt{cx}(0,2)$ share a target, so they commute; swap the adjacent pair in the middle. Then use the little identity (time order, T's on the target)

$$\mathtt{cx},\ T^\dagger,\ \mathtt{cx},\ T,\ \mathtt{cx} \;=\; T,\ \mathtt{cx},\ T^\dagger,$$

which you can check on a basis state $|b\rangle|c\rangle$: both sides map $c \mapsto b \oplus c$ and attach the phase $\omega^{\,c-(b\oplus c)}$. After that the sequence on qubit 2 is $\mathtt{cx}(1,2), T^\dagger, \mathtt{cx}(0,2), T, \mathtt{cx}(1,2), T^\dagger, \mathtt{cx}(0,2)$, four cx. With the two $\mathtt{cx}(0,1)$ we are at six, and this is the circuit everyone uses:



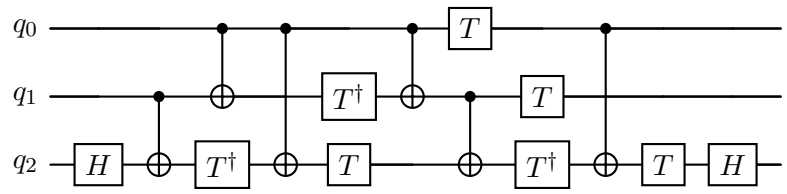
ccz with six cx and seven T-type gates (four T, three $T^\dagger$). No leftover phase.

There is a nice way to convince yourself this is right without any matrix algebra. Send in a basis state $|abc\rangle$. The cx's only ever XOR wire values, so each $T^{\pm 1}$ sits on a wire carrying one of a, b, c, $a \oplus b$, $a \oplus c$, $b \oplus c$, $a \oplus b \oplus c$ and contributes $\omega^{\pm(\text{that value})}$. Reading the circuit from left to right, the total power of ω is

$$k = a + b + c - (a \oplus b) - (a \oplus c) - (b \oplus c) + (a \oplus b \oplus c),$$

and the cx's all undo each other so the state comes out as $|abc\rangle$ again. For 000 every term vanishes. For a single 1, say 100: $k = 1 - 1 - 1 - 0 + 1 = 0$. For two 1's, say 110: $k = 2 - 0 - 1 - 1 + 0 = 0$. For 111: $k = 3 - 0 - 0 - 0 + 1 = 4$, and $\omega^4 = -1$. That is ccz.

By (b), sandwiching qubit 2 between two Hadamards turns this into the Toffoli:



The T-count is 7, and one cannot do better without extra qubits (that is a theorem, not something we can show here). Since in fault-tolerant architectures a T gate costs hundreds of physical operations while Cliffords are nearly free, people often quote the cost of a whole algorithm just as its T-count. Note also where the seven came from: `ccz` is the exponential of a cubic polynomial in the bits, and each monomial costs one $T^{\pm 1}$. The cubic monomial is the one that makes the gate non-Clifford.

Part C — What "universal" does and does not mean

6. Finite gate sets, counting, and the Cliffords

(a) A circuit built from $\mathcal{G}$ is a finite word $g_1 g_2 \cdots g_n$ in the letters of $\mathcal{G}$. There are $|\mathcal{G}|^n$ words of length n, finitely many, and the set of all words is the union of these over $n = 0, 1, 2, \ldots$, a countable union of finite sets, hence countable. (Different words can give the same matrix, but that only makes the set of circuits smaller.) The single-qubit gates, on the other hand, include $R_z(\theta)$ for every $\theta \in [0, 2\pi)$, all of them different matrices, so there are at least as many gates as there are points in an interval, which is uncountable. A countable set cannot cover an uncountable one, so some gate, in fact almost every gate, is not produced by any circuit. Exact universality is out of the question for a finite gate set, and from here on "universal" can only mean "approximable to any desired accuracy".

(b) Suppose $UXU^\dagger = VXV^\dagger$ and $UZU^\dagger = VZV^\dagger$, and set $W = V^\dagger U$. Multiplying the first equation by $V^\dagger$ on the left and V on the right gives $WXW^\dagger = X$, and likewise $WZW^\dagger = Z$. So W commutes with X and Z, hence with $Y = iXZ$, hence with every 2×2 matrix, since any such matrix is $m_0\mathbb{I} + m_1 X + m_2 Y + m_3 Z$. Only multiples of the identity commute with everything (if you prefer to see it directly: commuting with Z forces W to be diagonal, and commuting with X then forces its two diagonal entries to be equal). As W is unitary, $W = \mathrm{e}^{i\alpha}\mathbb{I}$, so $U = VW = \mathrm{e}^{i\alpha}V$. The two gates differ by a phase, i.e. they are the same gate.

(c) X has to go to a signed Pauli: six options, $\pm X, \pm Y, \pm Z$. Conjugation preserves algebraic relations, and $XZ = -ZX$, so the images of X and Z must anticommute. That rules out $\pm$(image of X) for the image of Z and leaves the other two Paulis with either sign: four options. By (b) the pair of images pins down the gate, and $Y = iXZ$ takes care of itself. So there are at most $6 \times 4 = 24$ Clifford gates up to phase.

I say "at most" because the argument does not show that every one of the 24 combinations is actually realised by some unitary. The honest way to finish is to write them all down, and (d) tells us where to look: products of H and S. Here is one word for each, sorted by length. The 24 pairs of images are all different, so these really are 24 different gates and the count is exactly 24.

	word	$X \mapsto$	$Z \mapsto$		word	$X \mapsto$	$Z \mapsto$
0	$\mathbb{I}$	X	Z	4	$HSSH$	X	$-Z$
1	H	Z	X	4	$SHSH$	Z	$-Y$
1	S	Y	Z	4	$SHSS$	Y	$-X$
2	HS	$-Y$	X	4	$SSHS$	$-Z$	Y
2	SH	Z	Y	4	$SSSH$	Y	X
2	SS	$-X$	Z	5	$HSSHS$	Y	$-Z$
3	HSH	X	$-Y$	5	$SHSSH$	$-Y$	$-Z$
3	HSS	Z	$-X$	5	$SSHSH$	$-X$	$-Y$
3	SHS	X	Y	5	$SSHSS$	$-Z$	$-X$
3	SSH	$-Z$	X	5	$SSSHS$	$-X$	Y
3	SSS	$-Y$	Z	6	$HSSHSH$	$-Y$	$-X$
				6	$SSHSSH$	$-X$	$-Z$
				6	$SSSHSH$	$-Z$	$-Y$

The 24 single-qubit Cliffords as words in H and S, leftmost letter acting last, with the number of letters in front.

If you like geometry, there is a quicker way to see 24: a Clifford permutes the six points $\pm\hat{x}, \pm\hat{y}, \pm\hat{z}$ on the Bloch sphere and keeps the handedness, so the Clifford group is the rotation group of a cube. A face can go to six places and there are four rotations fixing a face, $6 \times 4 = 24$.

(d) We are told that every circuit of H and S gates is one of the 24 gates in the table. The set of single-qubit gates is a continuum (that was (a)), and 24 points cannot be arbitrarily close to every point of a continuum. Pick any gate that is not in the table, say $R_z(\theta)$ with a small non-zero θ; its distance to the nearest of the 24 is some fixed positive number, and no Clifford circuit, however long, gets closer than that. Compare this with (a): there the problem was only exactness, and approximation got around it. Here the set of reachable gates does not grow at all with the length of the circuit, so approximation does not help either. The contrast with $\{H, T\}$ is the whole point of the sheet: one non-Clifford gate makes the reachable set infinite (Q3(d)) and then dense (Q3(e)).

The Remark on the sheet gives the other reason to care. By Gottesman–Knill, Clifford circuits can be simulated efficiently on a classical computer, so whatever makes quantum computers powerful, it is not in the Cliffords.

7. Errors add up

(a) Take any normalised $|\psi\rangle$. With the hint and the triangle inequality,

$$\begin{aligned}\|(U_2U_1 - V_2V_1)|\psi\rangle\| &= \|(U_2 - V_2)U_1|\psi\rangle + V_2(U_1 - V_1)|\psi\rangle\| \\ &\le \|(U_2 - V_2)\,U_1|\psi\rangle\| + \|V_2\,(U_1 - V_1)|\psi\rangle\|\,.\end{aligned}$$

In the first term $U_1|\psi\rangle$ is some normalised state, so the term is at most $E(U_2, V_2)$. In the second, V_2 is unitary and does not change norms, so the term is $\|(U_1 - V_1)|\psi\rangle\| \le E(U_1, V_1)$. Now maximise the left-hand side over $|\psi\rangle$.

(b) Apply (a) over and over (or do an induction on N if you want to be formal):

$$E(U_N \cdots U_1,\ V_N \cdots V_1) \le \sum_{i=1}^{N} E(U_i, V_i).$$

So accuracy ε/N per gate is enough for accuracy ε overall. Errors add, they do not multiply and they certainly do not grow exponentially. This is a worst-case bound; in practice errors partly cancel, but we are not allowed to count on that.

(c) Each of the N gates needs accuracy $\delta = \varepsilon/N$, which costs $O(\log^c(N/\varepsilon))$ gates from the universal set, so the total is

$$O\Big(N \log^c \frac{N}{\varepsilon}\Big).$$

The thing to notice is that the accuracy enters only through a logarithm. One more decimal place of accuracy costs an additive, not a multiplicative, amount per gate, and the dependence on N is essentially linear. To appreciate this, compare with what Q3(d) alone would give: the pigeonhole argument there only promises a suitable power k of order $1/\delta$, so the total cost would be about $N \cdot N/\varepsilon$, which is polynomial in $1/\varepsilon$ and quadratic in N. Solovay–Kitaev turns the $1/\delta$ into $\log^c(1/\delta)$, an exponential improvement, and that is what makes compiling into H, T and `cx` cheap enough to be practical rather than merely possible in principle.

8. On a computer (optional)

```
from qiskit import QuantumCircuit, transpile
from qiskit.quantum_info import Operator

# ccz from Q5(e); gates are listed in time order
qc1 = QuantumCircuit(3)
qc1.cx(1, 2); qc1.tdg(2)
qc1.cx(0, 1); qc1.cx(0, 2); qc1.t(2); qc1.tdg(1)
qc1.cx(0, 1); qc1.cx(1, 2); qc1.t(0); qc1.tdg(2); qc1.t(1)
qc1.cx(0, 2); qc1.t(2)

qc2 = QuantumCircuit(3); qc2.ccz(0, 1, 2)
print(Operator(qc1).equiv(Operator(qc2)))        # True

qc = QuantumCircuit(3); qc.ccx(0, 1, 2)
tqc = transpile(qc, basis_gates=['h', 't', 'tdg', 'cx'])
print(tqc.count_ops())
```

The first line prints `True`. The transpiled Toffoli comes out with 4 `t`, 3 `tdg`, 6 `cx` and 2 `h`, so Qiskit's own decomposition is the same circuit as in Q5(e), T-count 7. (Qiskit puts qubit 0 at the top of a drawing but at the right-hand end of a basis label; neither affects the counts.)